

PRACTICAL OSSP (WEEK 7)

SESSION 7

TASK 1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
koumudi@Koumudi: ~/OSSP/ × + ▾
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;
int global_uninitialized;

static int static_var = 20;
static int static_uninitialized;

void code_function()
{
    printf("This is a code segment function.\n");
}

int main()
{
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 40;

    printf("Linux Process Memory Address Space\n");
    printf("-----\n");

    printf("Code    variable/function address : %p\n", (void *)code_function);
    printf("Global initialized variable      : %p\n", (void *)&global_var);
    printf("Global uninitialized variable     : %p\n", (void *)&global_uninitialized);
    printf("Static initialized variable       : %p\n", (void *)&static_var);
    printf("Static uninitialized variable     : %p\n", (void *)&static_uninitialized);
    printf("Heap variable                     : %p\n", (void *)heap_var);
    printf("Stack variable                    : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

```
koumudi@Koumudi: ~/OSSP/ × + ▾
koumudi@Koumudi:~$ cd ~/OSSP/PRACTICAL
koumudi@Koumudi:~/OSSP/PRACTICAL$ mkdir Session_7
koumudi@Koumudi:~/OSSP/PRACTICAL$ cd Session_7
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ pwd
/home/koumudi/OSSP/PRACTICAL/Session_7
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ ls
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ nano memory_segments.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ gcc memory_segments.c -o memory_segments
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ ls
memory_segments  memory_segments.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ ./memory_segments
Linux Process Memory Address Space
-----
Code    variable/function address : 0x5e8d828071e9
Global initialized variable      : 0x5e8d8280a010
Global uninitialized variable     : 0x5e8d8280a01c
Static initialized variable       : 0x5e8d8280a014
Static uninitialized variable     : 0x5e8d8280a020
Heap variable                     : 0x5e8db57b62a0
Stack variable                    : 0x7ffe582edf0c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ |
```

TASK 2: Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
koumudi@Koumudi: ~/OSSP/ × + ∨
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
static int static_var = 200;

int main()
{
    int stack_var = 300;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 400;

    printf("Process ID: %d\n", getpid());

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address : %p\n", (void *)heap_var);
    printf("Stack address : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to terminate.\n");

    while (1)
    {
        sleep(1);
    }

    free(heap_var);

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ nano memory_maps.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$
-bash: koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$: No such file or directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ gcc memory_maps.c -o memory_maps
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ ls
memory_maps  memory_maps.c  memory_segments  memory_segments.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ ./memory_maps
Process ID: 467
Global address : 0x594acc6ae010
Static address : 0x594acc6ae014
Heap address : 0x594ade9d82a0
Stack address : 0x7ffee4a2d10c

Process is running...
Press Ctrl+C to terminate.
```

```
koumudi@Koumudi: ~/OSSP/ × + v
koumudi@Koumudi:~$ cd ~/OSSP/PRACTICAL/Session_7
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ cat /proc/467/maps
594acc6aa000-594acc6ab000 r--p 00000000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ab000-594acc6ac000 r-xp 00001000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ac000-594acc6ad000 r--p 00002000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ad000-594acc6ae000 r--p 00002000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ae000-594acc6af000 rw-p 00003000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594ade9d8000-594ade9f9000 rw-p 00000000 00:00 0 [heap]
777a3ea00000-777a3ea28000 r--p 00000000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ea28000-777a3ebb1000 r-xp 00028000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ebb1000-777a3ec00000 r--p 001b1000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec00000-777a3ec04000 r--p 001ff000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec04000-777a3ec06000 rw-p 00203000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec06000-777a3ec13000 rw-p 00000000 00:00 0
777a3ed15000-777a3ed18000 rw-p 00000000 00:00 0
777a3ed1d000-777a3ed1f000 rw-p 00000000 00:00 0
777a3ed1f000-777a3ed23000 r--p 00000000 00:00 0 [vvar]
777a3ed23000-777a3ed25000 r--p 00000000 00:00 0 [vvar_vclock]
777a3ed25000-777a3ed27000 r-xp 00000000 00:00 0 [vdso]
777a3ed27000-777a3ed28000 r--p 00000000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed28000-777a3ed53000 r-xp 00001000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed53000-777a3ed50000 r--p 0002c000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed50000-777a3ed5f000 r--p 00036000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed5f000-777a3ed61000 rw-p 00038000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffee4a0e000-7ffee4a2f000 rw-p 00000000 00:00 0 [stack]
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ pmap 467
467: /memory_maps
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ pmap 467
467: /memory_maps
0000594acc6aa000 4K r---- memory_maps
0000594acc6ab000 4K r-x-- memory_maps
0000594acc6ac000 4K r---- memory_maps
0000594acc6ad000 4K r---- memory_maps
0000594acc6ae000 4K rw--- memory_maps
0000594ade9d8000 132K rw--- [ anon ]
0000777a3ea00000 160K r---- libc.so.6
0000777a3ea28000 1572K r-x-- libc.so.6
0000777a3ebb1000 316K r---- libc.so.6
0000777a3ec00000 16K r---- libc.so.6
0000777a3ec04000 8K rw--- libc.so.6
0000777a3ec06000 52K rw--- [ anon ]
0000777a3ed15000 12K rw--- [ anon ]
0000777a3ed1d000 8K rw--- [ anon ]
0000777a3ed1f000 16K r---- [ anon ]
0000777a3ed23000 8K r---- [ anon ]
0000777a3ed25000 8K r-x-- [ anon ]
0000777a3ed27000 4K r---- ld-linux-x86-64.so.2
0000777a3ed28000 172K r-x-- ld-linux-x86-64.so.2
0000777a3ed53000 40K r---- ld-linux-x86-64.so.2
0000777a3ed5d000 8K r---- ld-linux-x86-64.so.2
0000777a3ed5f000 8K rw--- ld-linux-x86-64.so.2
00007ffee4a0e000 132K rw--- [ stack ]
total 2692K
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ cat /proc/467/maps
```

```
total 2692K
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ cat /proc/467/maps
594acc6aa000-594acc6ab000 r--p 00000000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ab000-594acc6ac000 r-xp 00001000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ac000-594acc6ad000 r--p 00002000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ad000-594acc6ae000 r--p 00002000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594acc6ae000-594acc6af000 rw-p 00003000 08:30 11790 /home/koumudi/OSSP/PRACTICAL/Session_7/memory_maps
594ade9d8000-594ade9f9000 rw-p 00000000 00:00 0 [heap]
777a3ea00000-777a3ea28000 r--p 00000000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ea28000-777a3ebb1000 r-xp 00028000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ebb1000-777a3ec00000 r--p 001b1000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec00000-777a3ec04000 r--p 001ff000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec04000-777a3ec06000 rw-p 00203000 08:30 46258 /usr/lib/x86_64-linux-gnu/libc.so.6
777a3ec06000-777a3ec13000 rw-p 00000000 00:00 0
777a3ed15000-777a3ed18000 rw-p 00000000 00:00 0
777a3ed1d000-777a3ed1f000 rw-p 00000000 00:00 0
777a3ed1f000-777a3ed23000 r--p 00000000 00:00 0 [vvar]
777a3ed23000-777a3ed25000 r--p 00000000 00:00 0 [vvar_vclock]
777a3ed25000-777a3ed27000 r-xp 00000000 00:00 0 [vdso]
777a3ed27000-777a3ed28000 r--p 00000000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed28000-777a3ed53000 r-xp 00001000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed53000-777a3ed5d000 r--p 0002c000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed5d000-777a3ed5f000 r--p 00036000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
777a3ed5f000-777a3ed61000 rw-p 00038000 08:30 46251 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffee4a0e000-7ffee4a2f000 rw-p 00000000 00:00 0 [stack]
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_7$ cat /proc/467/status
```

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```
7ffee4a0e000-7ffee4a2f000 rw-p 00000000 00:00 0
kroumudi@Kroumudi:~/OSSP/PRACTICAL/Session_7$ cat /proc/467/status
Name:      memory_maps
Umask:     0022
State:     S (sleeping)
Tgid:      467
Ngid:      0
Pid:       467
PPid:      300
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NSTgid:    467
NSpid:     467
NSpgid:    467
NSSid:     300
Kthreadd:  0
VmPeak:    2692 kB
VmSize:    2692 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     1736 kB
VmRSS:     1736 kB
RssAnon:           100 kB
RssFile:          1636 kB
RssShmem:          0 kB
VmData:      224 kB
VmStk:       132 kB
VmExe:        4 kB
VmLib:      1752 kB
VmPTE:       44 kB
VmSwap:       0 kB
HugetlbPages:      0 kB
CoreDumppg: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:     1
SigQ:       1/31095
SigPnd:    0000000000000000
ShdPnd:    0000000000000000
SigBlk:    0000000000000000
SigIgn:    0000000000000000
SigCgt:    0000000000000000
CapInh:    0000000000000000
```

```
capAmb: 0000000000000000
CapAmb: 0000000000000000
NoNewPrivs: 0
Seccomp: 0
Secomp_filters: 0
Speculation_Store_Bypass: thread vulnerable
SpeculationIndirectBranch: conditional enabled
Cpus_allowed: fff
Cpus_allowed_list: 0-11
Mems_allowed_list: 0
voluntary_ctx_switches: 188
nonvoluntary_ctx_switches: 0
673 434 409 1 0 89 0
```